

IN-CHANNEL HYDRAULIC STRUCTURES

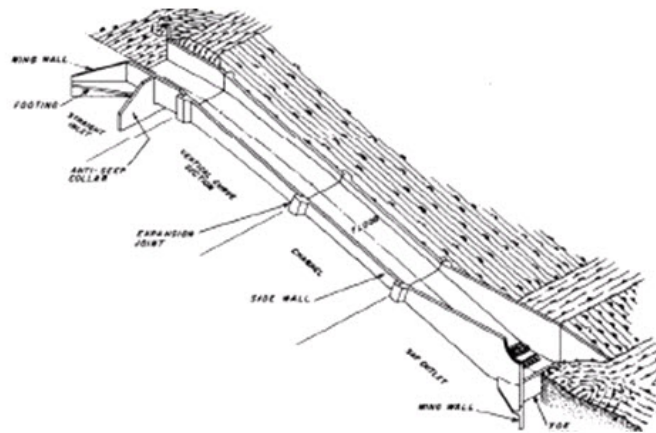


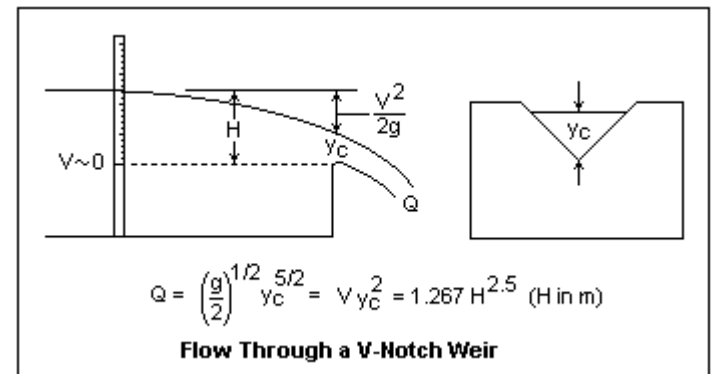
Fig. 13.5. Chute structures. (Source: SCS, 1984)

In-channel Hydraulic structures

- Interventions and devices in streams/channels/conduits
 - measure,
 - regulate or
 - discharge flows.
- This subject covers hydraulic structures relating to irrigation, drainage, water supply, and hydro power projects.

Hydraulic Structures

- Weirs, culverts, flumes, etc.



Planning and Design Considerations

Planning

- Identify need for hydraulic structure
- Preliminary type and site selection
- Collection of existing data
- Preliminary design

Detailed Design

- Final site selection
- Estimate area required for structure and ancillary works
- Additional data collection

Steps in Detailed Design

- Hydrologic design
- Hydraulic design
- Geotechnical design
- Structural design
- Detail costing, specifications, and other documentation

Hydrologic design

- Runoff
 - Hydrographs
 - Flow rates
 - streams
 - Modelling runoff
 - Rational method
 - SCS method
 - Other methods...
 - HEC-HMS !!
 - Problems?
 - ground flow
 - drainage

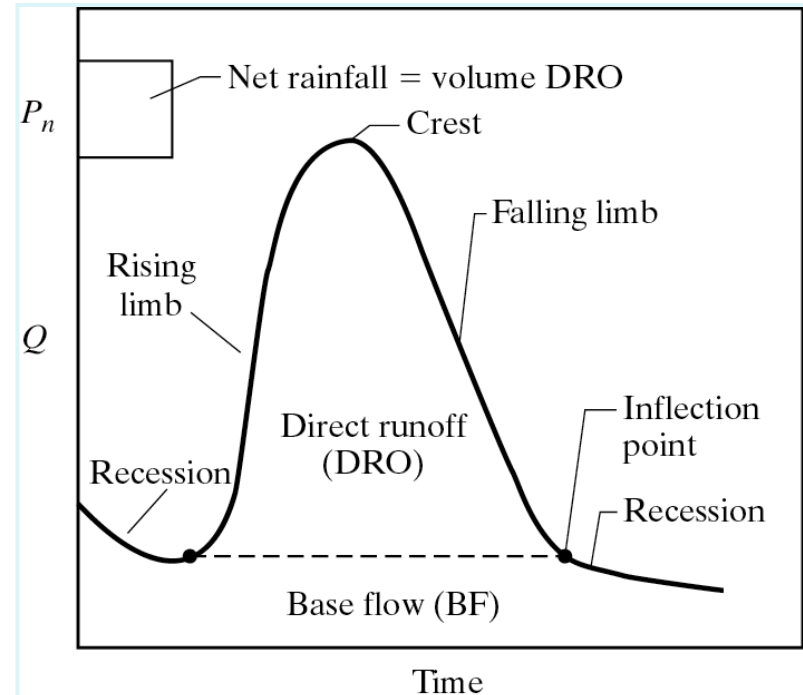


Figure 2.3
Hydrograph relations.

Hydrologic design

- Land use change
 - Design for future scenarios
- Climate change

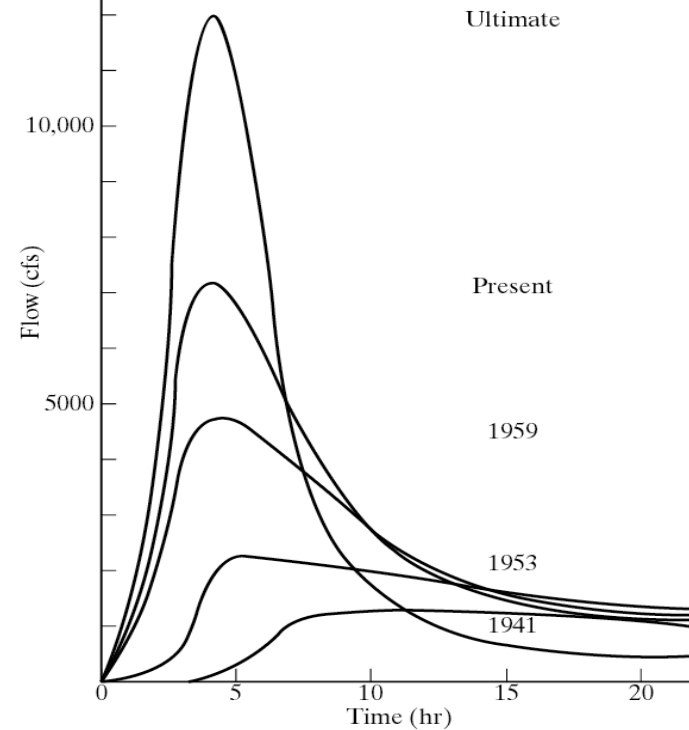
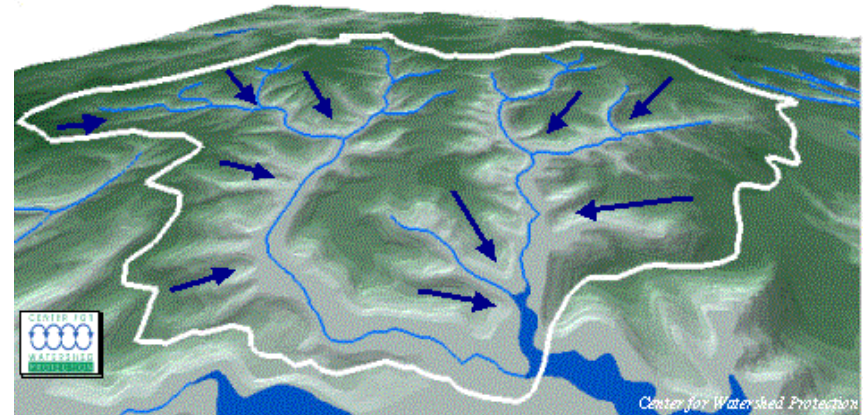


Figure 2.12

Brays Bayou unit hydrograph changes as a function of land use changes.

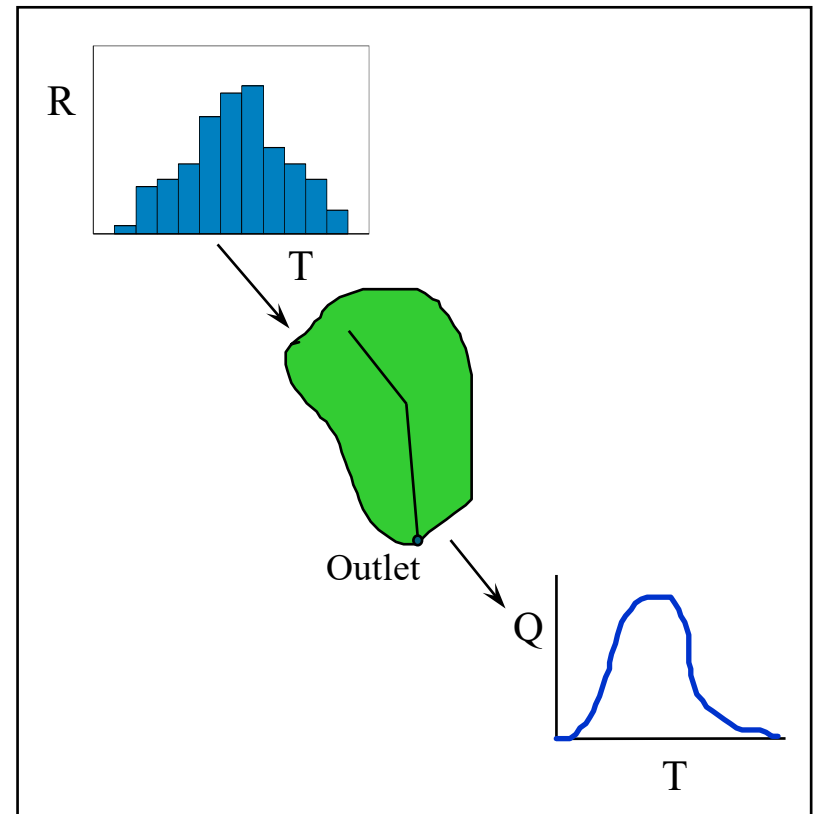
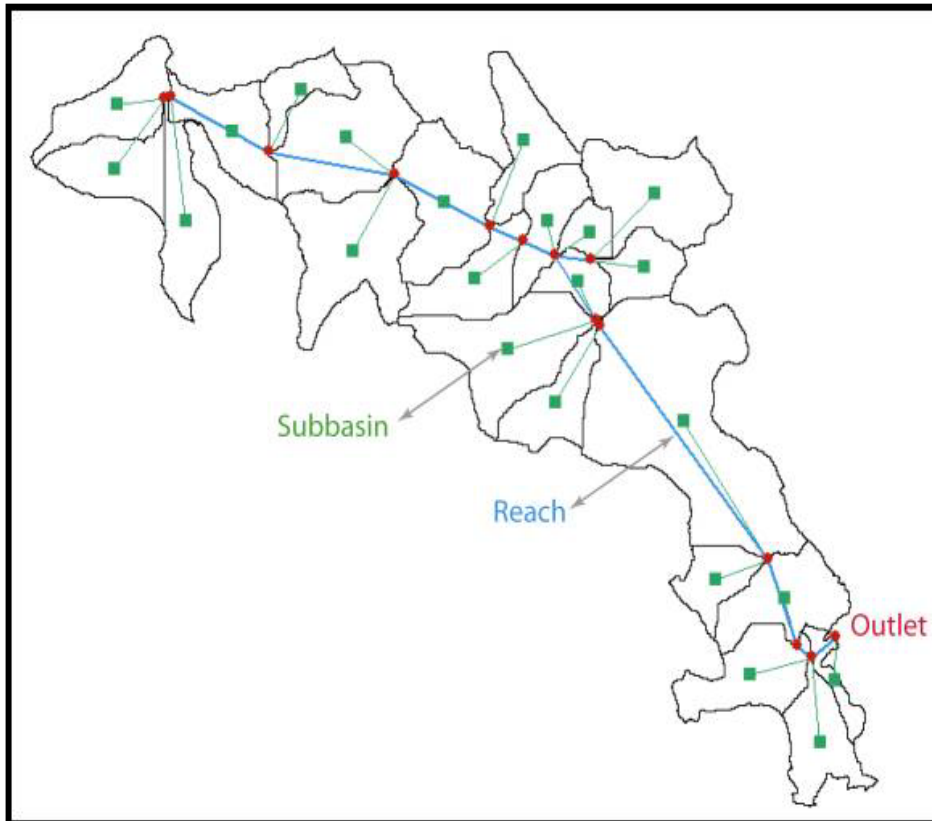
Hydrologic design

- Catchments (design flows)
 - Peak flows
 - Low flows
 - Durations (T_c , T_p , etc.)
- Erosion? Sediment transport?
- Using GIS?
Information / data..



GIS + hydrologic models (ex. HEC-HMS)

- Converts rain to streamflow



Hydraulic design

- Water levels
- Flow rates
- Energy losses

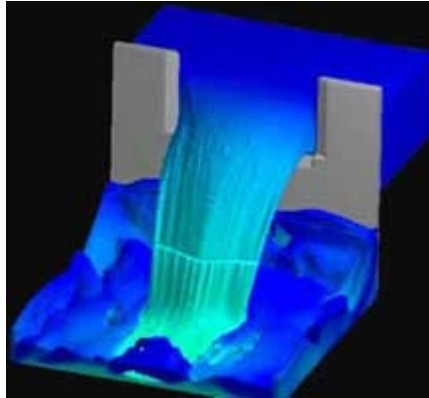
- Manual calculations
- Specific programs
 - HEC-RAS, Aulos, Mike 11, others..

Hydraulic design

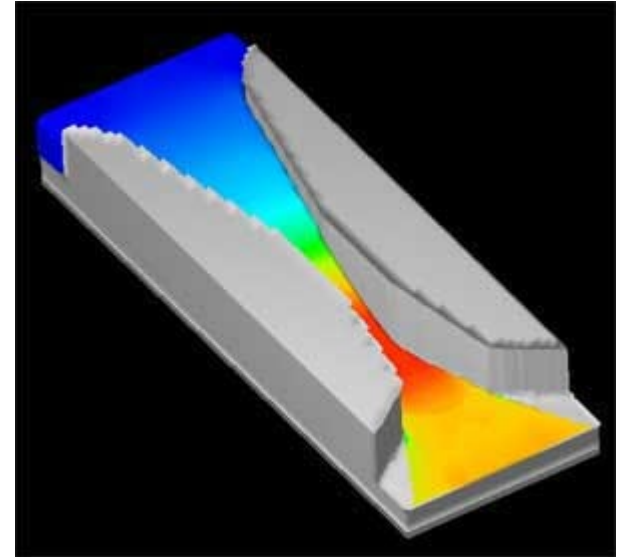
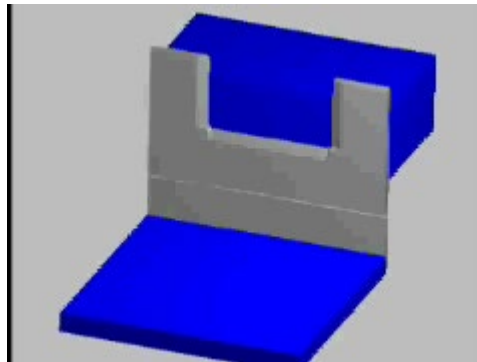
- Functional requirements of hydraulic design:
 - Sufficient capacity to pass the design discharge
 - Adequate overflow capacity (in case of extreme events or miss operation)
 - Kinetic energy of the discharge must be dissipated within the confines of the structure
 - Protect structure
 - Protect downstream channel from damage or erosion
 - Minimize hydraulic loss in certain transitions by correct structural proportioning

Hydraulic models

- 1 D models
- 2 D models
- 3 D models



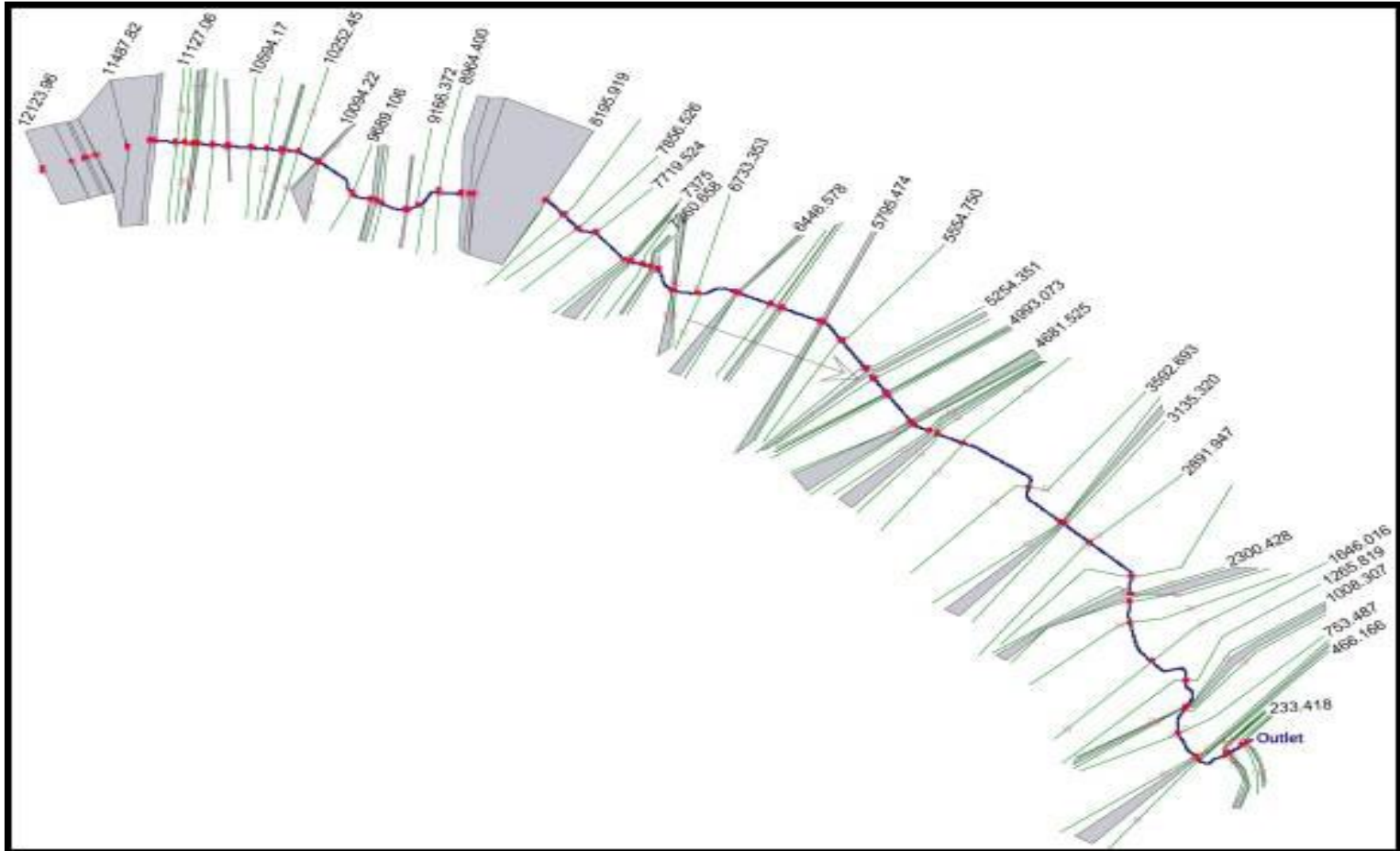
Weir



Parshall Flume

HEC-RAS, Aulos, Mike 11, etc

- Converts streamflows to water surface elevations



Hydraulic design

- Primary causes of failure of channels and in-channel hydraulic structures:
 - Insufficient hydraulic capacity
 - Insufficient provision for energy dissipation

Basic components of a hydraulic structure

- Inlet
- Conduit
- Outlet

Hydraulic Design

- Hydraulic control – determine the location of the water surface control.
 - Downstream or upstream?
- Hydraulic head losses
 - Friction (Mannings formula – for concrete canal structure $n=0.014$, etc..)
 - Transitions and bends
- Discharge coefficients
 - Orifice
 - Weirs..

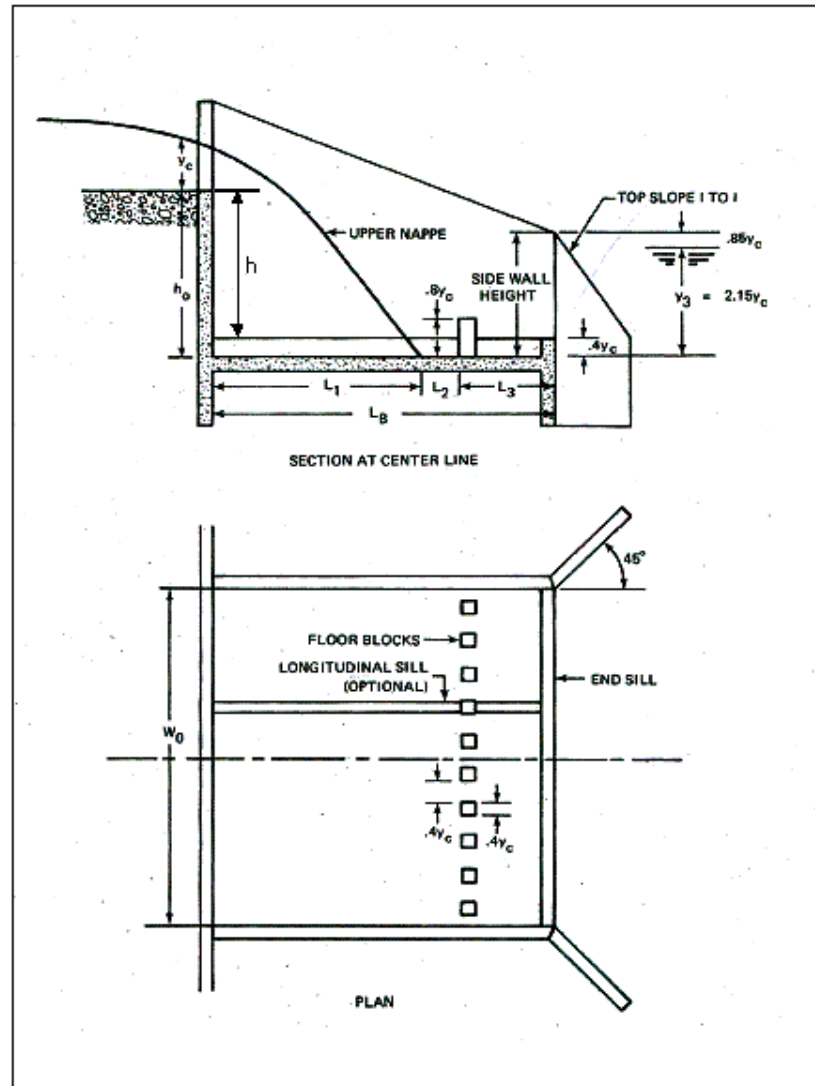
Geotechnical design

- Soils!!! (in this case must remove topsoil and organic mater)
- Stable foundation design
 - Resist sliding and overturning
 - Prevent percolating water from removing foundation materials
 - Provide foundation pressures less than max. allowable bearing pressures
 - Seepage (around structure or underneath structure) may cause failure
- Scouring... erosion...

Structural design

- Materials, stresses, etc...
- Ease of construction
 - Curves?
 - Straight lines
 - Complexity
- Apart from hydraulic features, structure must include these feature (to prevent seepage under or around structure and prevent local erosion):
 - Suitable wing walls
 - Side walls,
 - Head wall extensions
 - Toe walls, etc...

- Side walls
- Wing walls
- Toe walls
- Head wall extensions



Straight drop structure

<http://www.fhwa.dot.gov/engineering/hydraulics/pubs/06086/hech14ch11.cfm>

Structural design

- Loads:
 - Dead load weights:
 - Water
 - Backfill
 - Concrete
 - Operating deck loads
 - Lateral pressures
 - Water
 - Earth
 - Construction and operating equipment wheel surcharge (wall have to withstand construction loads – wheel pressures transmitted through the earth)
 - Ice?
 - Other pressures:
 - Uplift – water percolating
 - Hydraulic transients – usually insignificant
 - Seismic
 - Wheel loads for buried pipes, culverts, etc..

Detail costing, specifications and other documentation

- Environmental Impact, RMA, etc..
- Available Budget
- Feed back loop – redesign?

Example: establishing need..

- Dimensions of a stable channel have been shown to be related to both discharge and soil type. **Site specific equations** are often developed to help engineers.
- Here is an example of **site specific equations** for a stable channel excavated through:

Clayey soil:

$$W = 3.0 Q^{1/2}$$

$$D = 0.75 Q^{1/3}$$

Sandy soil:

$$W = 4.0 Q^{1/2}$$

$$D = 0.6 Q^{1/3}$$

* Use Manning's to find required slope: $S = v^2 n^2 / r^{4/3}$

W – average channel width

D – flow depth

Q – discharge

b (bottom width) = $W - ZD$

Z is side slope (H:V)

$P = b + 2D(1 + z^2)^{1/2}$

$R = DW/P$

Example

- A ditch to convey $1.4 \text{ m}^3/\text{s}$ through a 7 km reach of clay type soil must be designed. The natural ground slope in the direction of the flow is 0.001. Approximately how many drop structures will be required in this reach?
- How about for a sandy soil?